Units 9 and 10

THE FINANCIAL SYSTEM

A. Income, Borrowing, and Saving

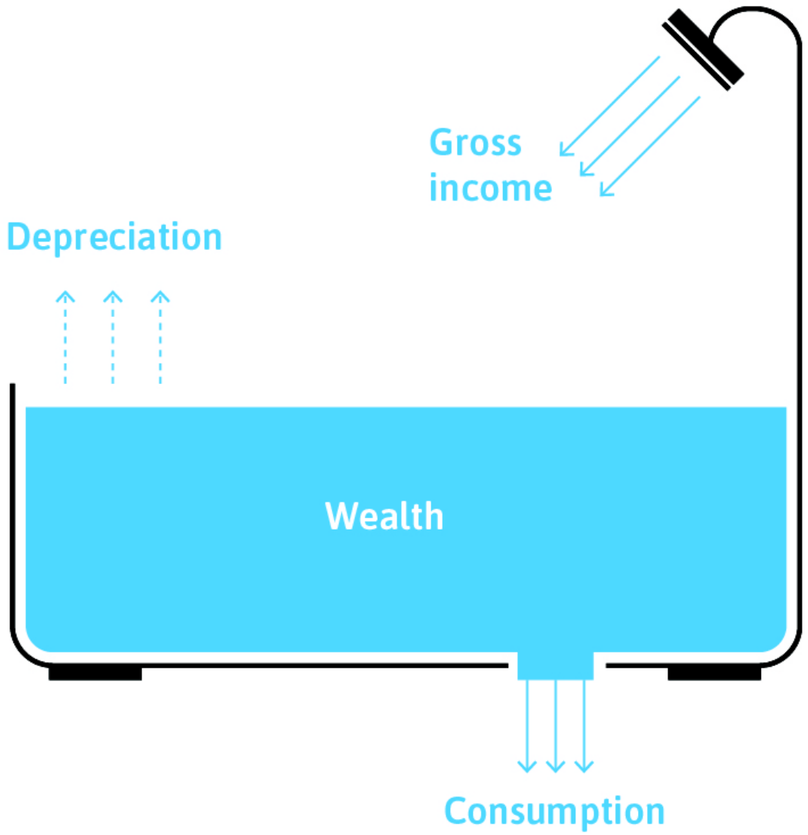
Income vs. Wealth

Wealth = stock of things owned or value of that stock.

= Buildings, land, machinery, capital goods – debts owed + debts owed to you

Income = the amount of money one receives over some period of time (flow).

- From market earnings, investments, government, etc.



Other key concepts

Depreciation = reduction in the value of a stock of wealth over time.

Net income = the maximum amount that one could consume without running down wealth.

Net income = gross income – depreciation

Savings = income that is not consumed

Investment = expenditure on newly produced capital goods

Consumption over time

There is a trade-off between consuming goods now and later.

The opportunity cost of having more goods now is having fewer goods later.

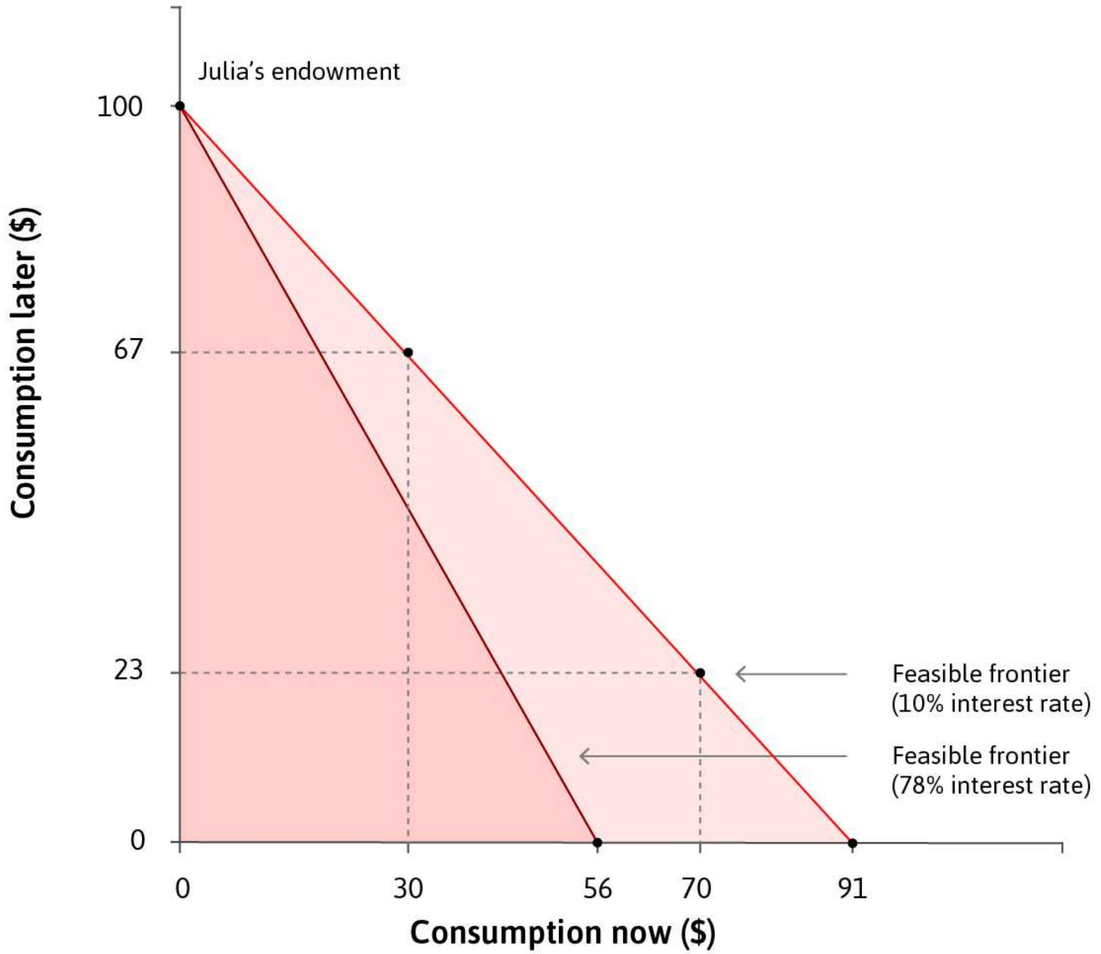
Borrowing and lending allow us to rearrange our capacity to buy goods and services across time

Borrowing

Borrowing allows us to buy more now, at the cost of buying less later.

Interest rate (**r**) = The price of bringing some buying power forward in time.

(1+r) = Trade-off between current and future consumption (MRT)



Preferences for consumption

Borrowing allows us to bring consumption forward

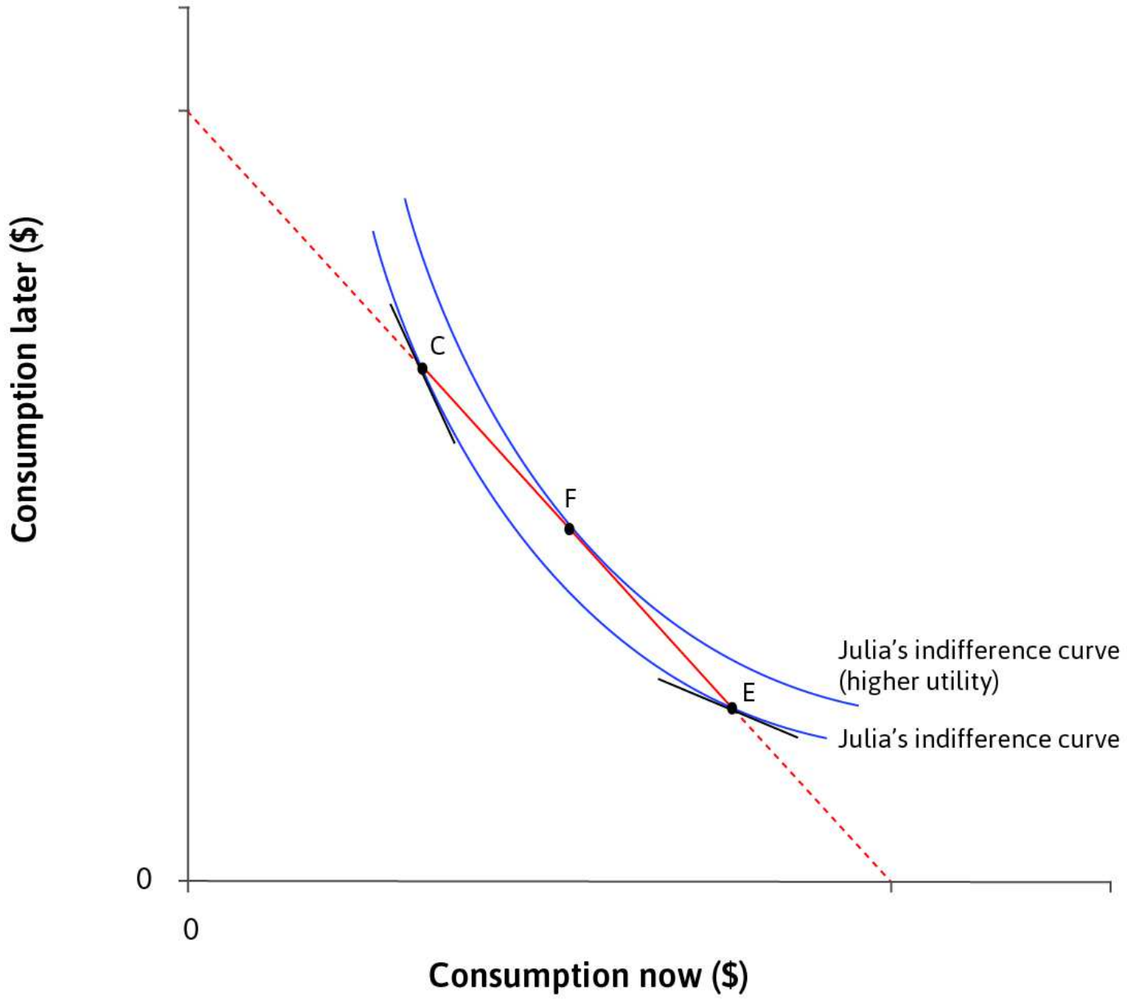
How much consumption an individual will bring forward depends on:

- Preferences for **consumption smoothing**
- **Impatience**

Consumption smoothing

Diminishing marginal returns to consumption = The value of an additional unit of consumption declines, the more consumption the individual has.

An individual **smoothes their consumption** to avoid consuming a lot in one period and little in the other.



Pure impatience

Consumption smoothing may seem like impatience

However, we differentiate it from **pure impatience** = favoring present consumption over future consumption

- Myopia : People can experience present satisfaction more strongly than the same satisfaction later
- Prudence: People know that they may not be around in the future, and so they want to consume now

Optimal decision-making

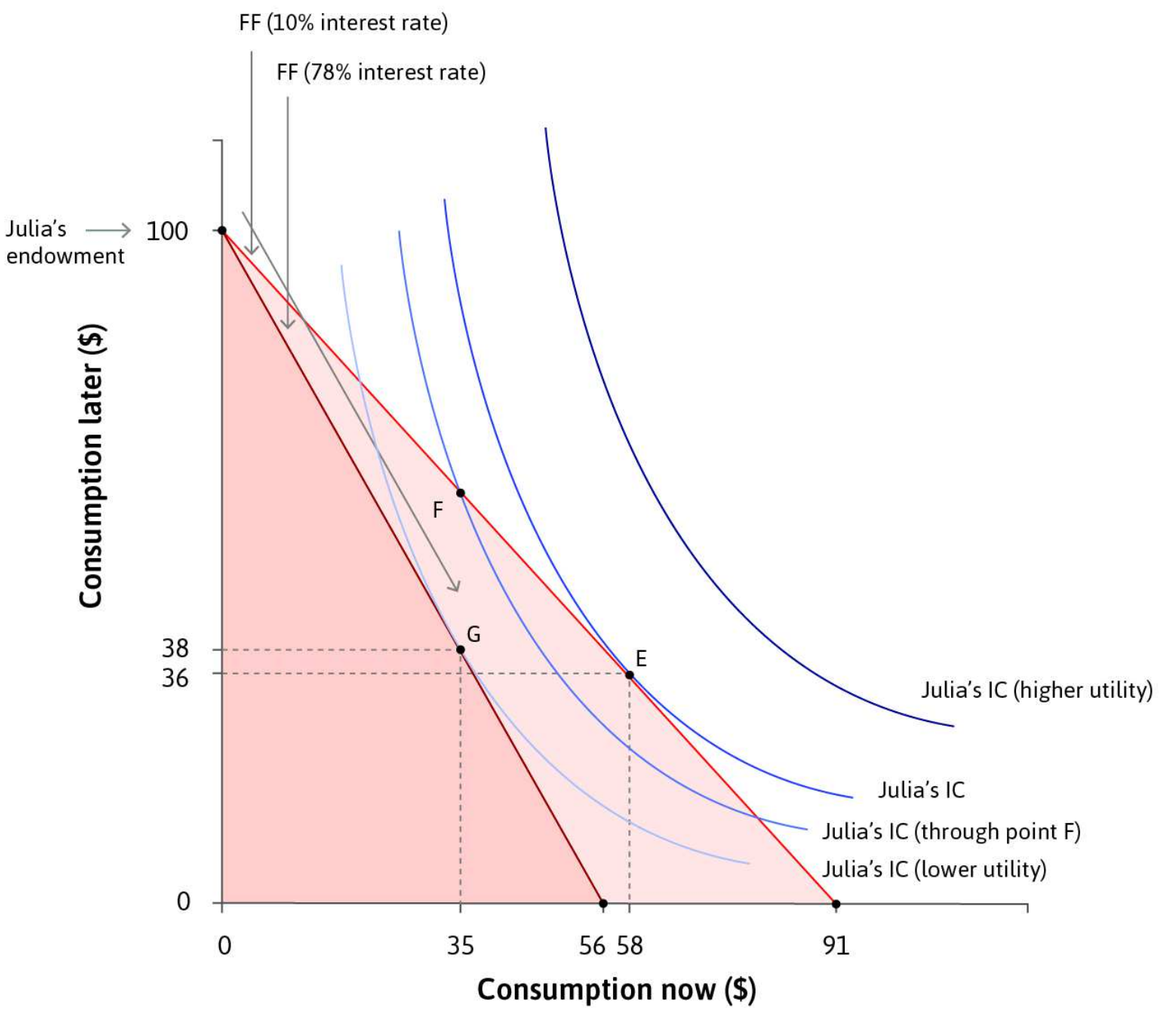
Discount rate (ρ) = a measure of a person's impatience.

IC are shifted toward present

Individual borrows where discount rate = interest rate

MRS = MRT

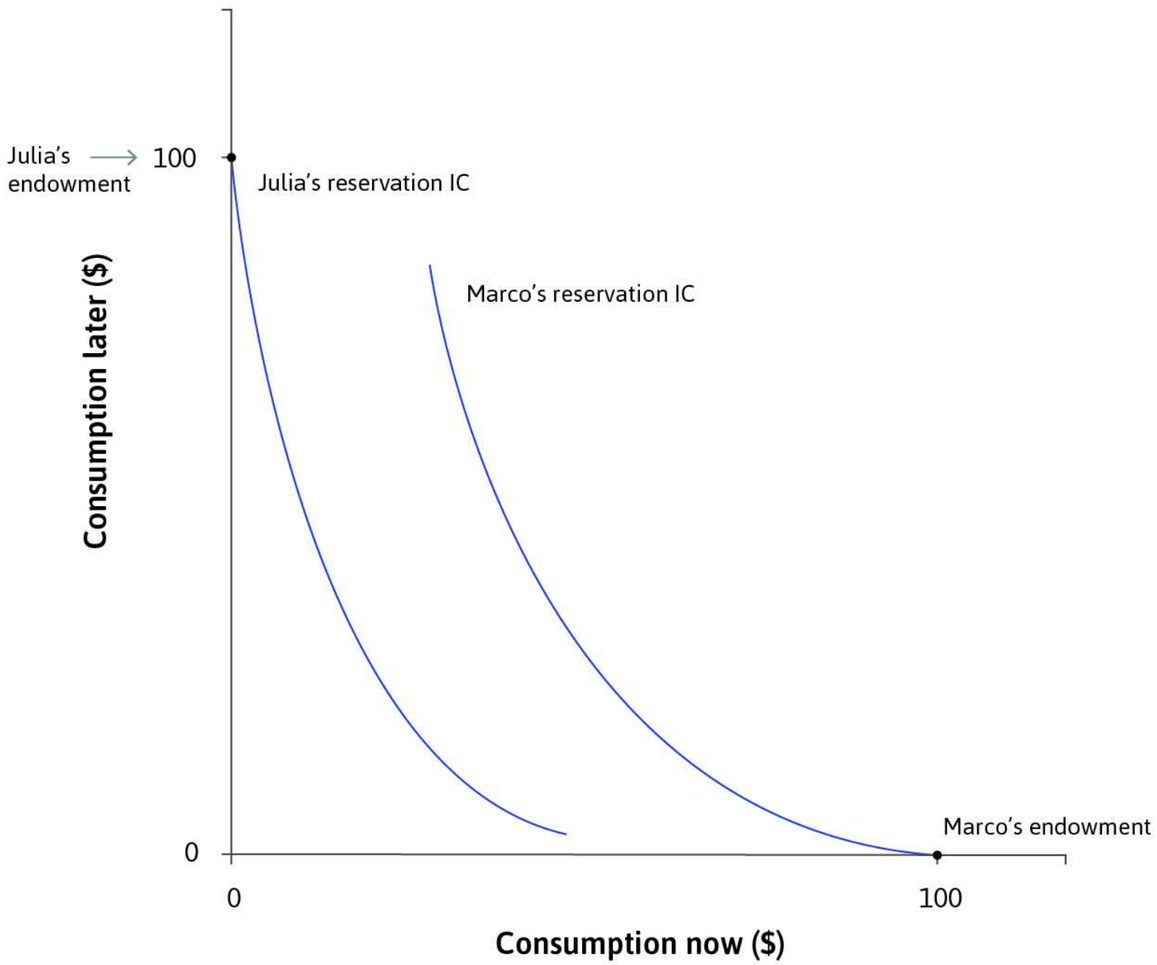
$1+\rho = 1+r$



Borrowers and savers

Reservation indifference curve = all of the points at which the individual would be just as well off as at the reservation position (endowment).

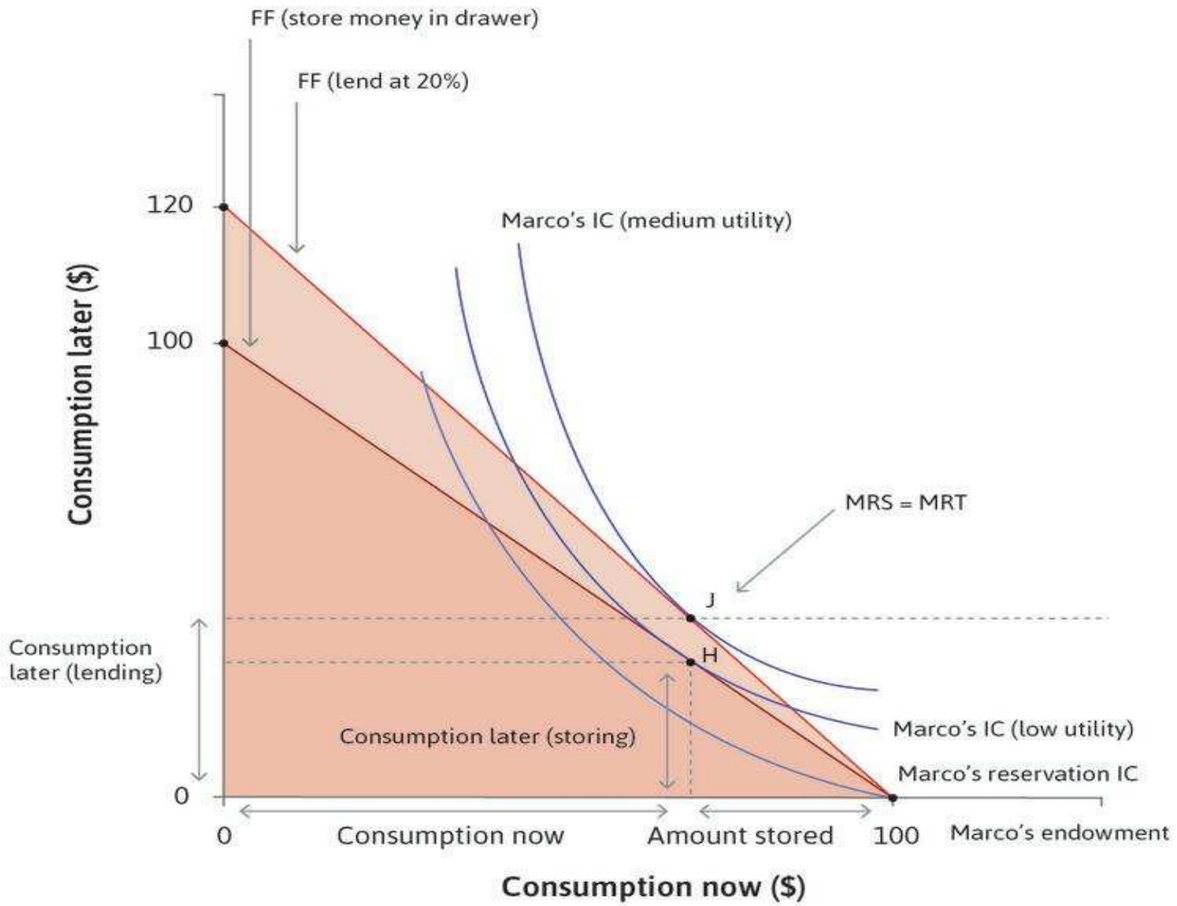
The borrower and the saver have different indifference curves because they have <u>different endowments</u>.



Saving and lending

A **saver** smoothes his consumption by postponing it into the future.

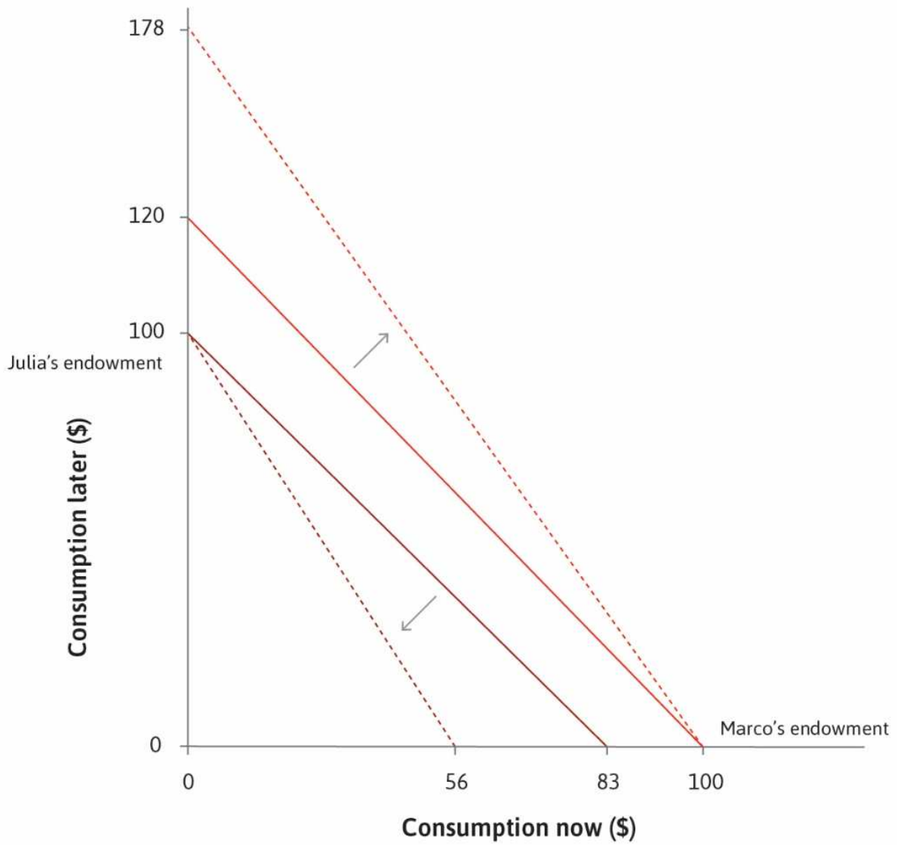
Lending money with interest <u>expands the saver's feasible set</u>, compared to simply storing it.



Conflict of Interest

When there is an an increase in the rate of interest:

- It improves **lender's** choices
- It shrinks **borrower's** choices



There is a **conflict of interest** over how the mutual gains from exchange are shared.

Conflict of interest = in order for one party to gain more from the interaction, another party must do less well.

B. Asymmetric Information in Lending

Principal-Agent Problem

Principal-agent problem =
a conflict of interest between principal and agent, about some hidden action or attribute of the agent that cannot be enforced or guaranteed in a binding contract.

E.g. Financing a project
Lenders face the risk that money borrowed will not be repaid, but lack information about the project's success or borrower's effort so cannot ensure that the project succeeds

Equity and Collateral

To resolve the conflict of interest between the principal (lender) and the agent (borrower):

- **Equity**: The lender may require the borrower to risk some of their own wealth on the project

- **Collateral**: The borrower has to set aside assets that will be transferred to the lender if the loan is not repaid

Credit Rationing

Those with less wealth find it more difficult to provide equity or collateral:

Credit rationing = when those with less wealth:

- borrow on unfavorable terms compared with those with more wealth (**credit-constrained**)
- or are refused loans entirely (**credit-excluded**)

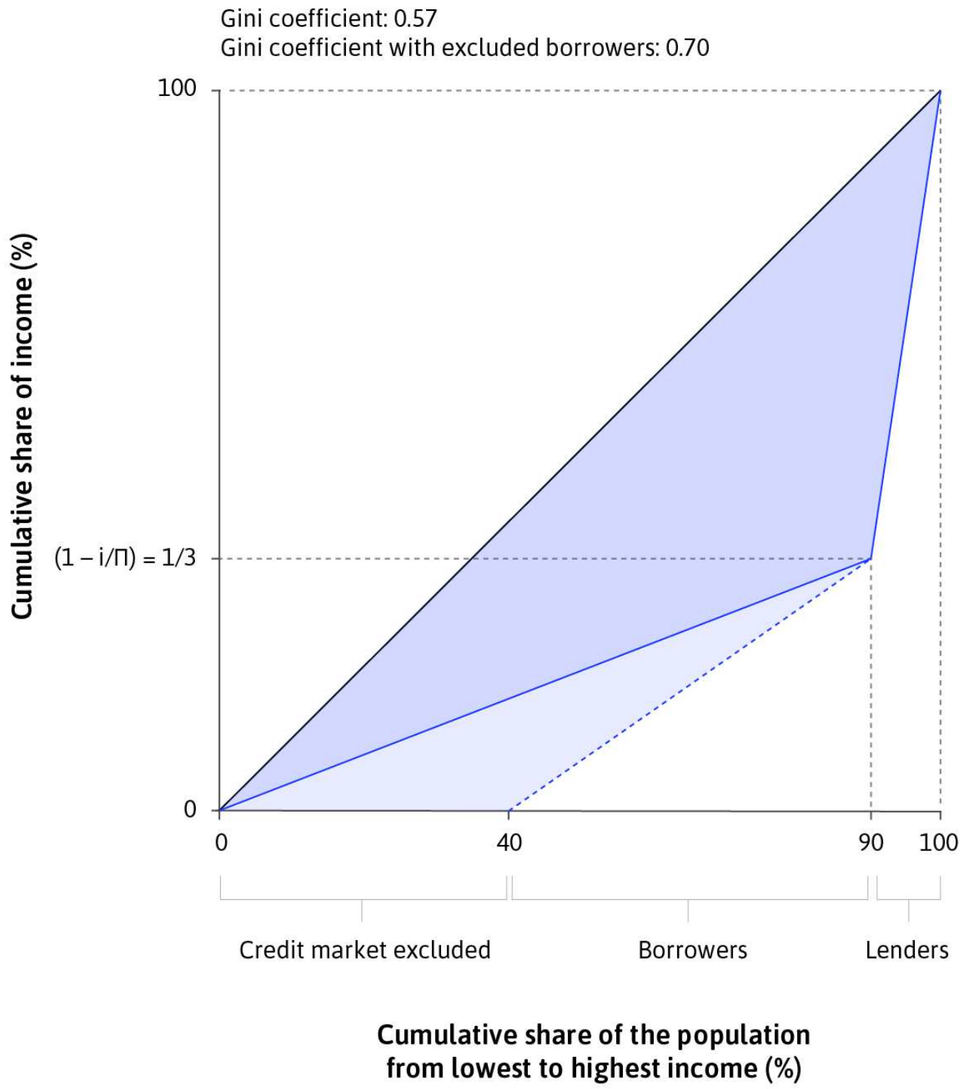
This is sometimes called a *net worth bias* in lending

C. Credit and Inequality

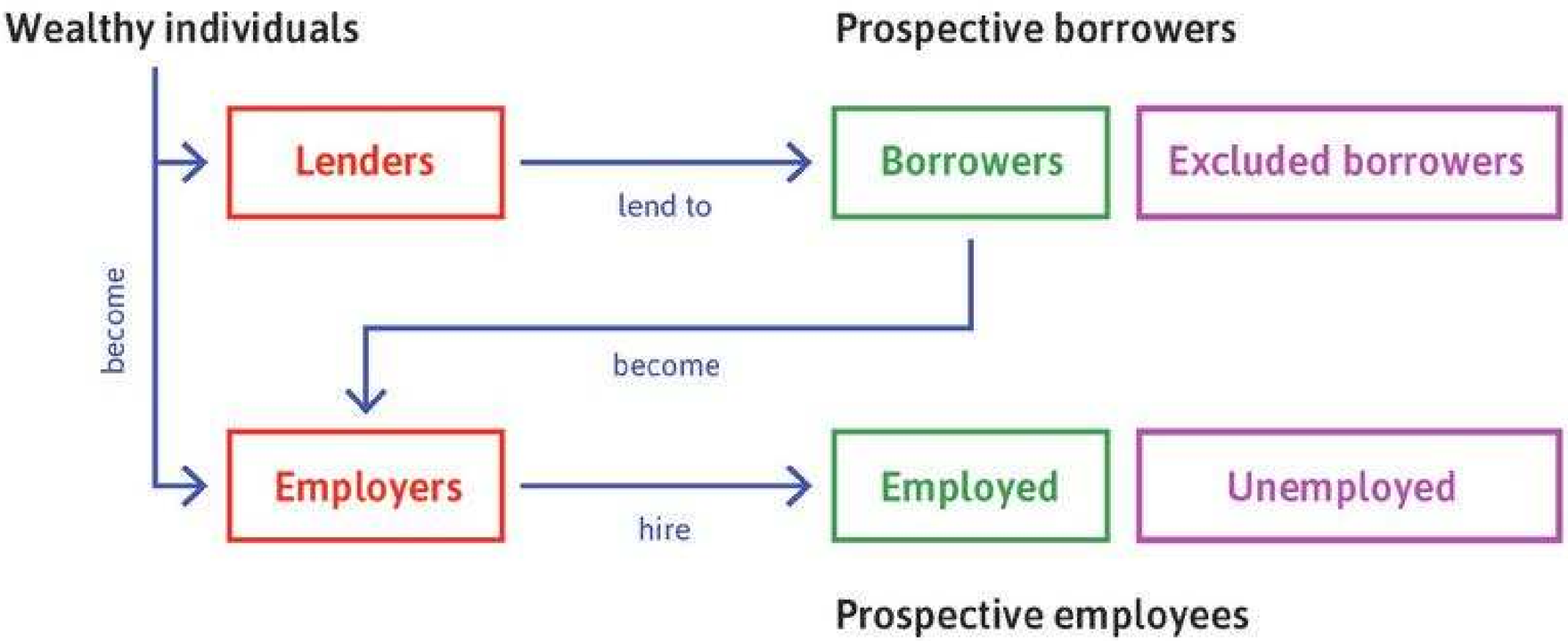
Inequality in Credit Access

Inequality may increase when some people are in a position to profit by lending money to others.

Credit-rationing increases inequality: people with limited wealth are not able to profit from investment opportunities that are open to those with more assets



Credit market and Labor market



Principals → Lenders and Employers
Agent → Borrowers and Employees

Principals have **power** over agents by <u>refusing to deal with them.</u>

D. Banks and Money

Banks

A **bank** is a firm that makes profits by lending and borrowing.

Banks borrow from households (**deposits**).

The interest they pay on deposits is lower than the interest they charge on loans, which is how banks make a profit

Money

Money fulfills these functions:

- A medium of exchange
- A store of value
- A unit of account

Money can be made from a commodity (gold, silver) or creating liabilities (commercial banks - bank accounts, central banks - coins and notes)

Central bank

Base money = notes and coins. Money as legal tender.

Legal tender has to be accepted as payment by law.

The **central bank** is the only bank that can create legal tender.

- The central bank is usually owned by the government
- It acts as the banker for the commercial banks, who have accounts at the central bank that hold legal tender
- By crediting these accounts, the central bank creates money

Bank money

Commercial banks create money by making loans

- This is called **bank money** ≠ legal tender
- It is a liability to the bank, not an asset
- Banks earn profits by charging interest on bank money

Bonus Bank gives Gino a loan of $100

Bonus Bank's assets	Bonus Bank's liabilities
$20 base money $100 bank loan ***Total: $120***	$120 payable on demand to Gino

Broad money = base money + bank money

Default risk and Liquidity risk

Banks provide the service of <u>maturity transformation:</u>

- Deposits can be withdrawn at any time…
- But loans only need to be repaid after a specified time.

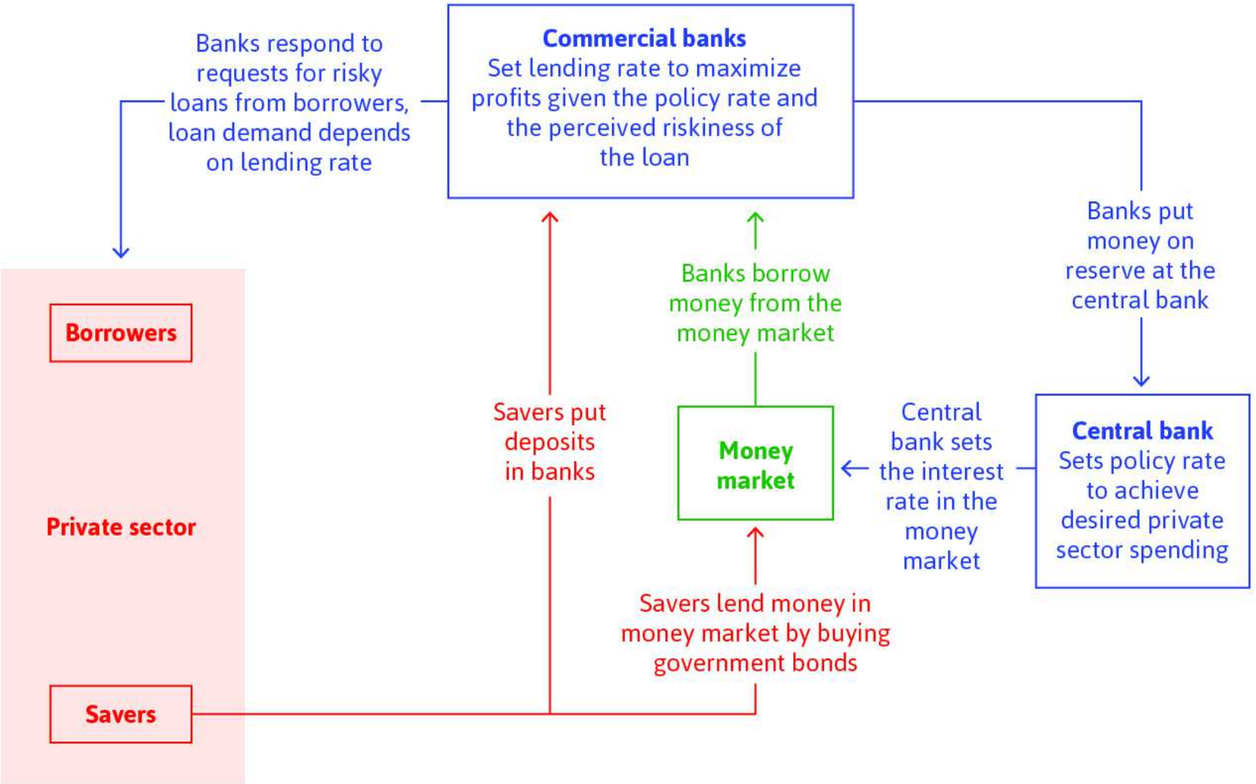
This is also <u>liquidity transformation</u>:

- Deposits are **liquid** (immediately usable as money)
- Value of loans given to borrowers are *illiquid*

This exposes the bank to risks:

1. **Default risk** (loans not repaid)
2. **Liquidity risk** (deposit demands exceed loan payments)

The Financial System



Policy interest rate = the interest rate on base money set by the central bank.

Bank lending rate = the average interest rate charged by commercial banks to firms and households.

E. Asset Price Bubbles

Asset Value

People buy assets for two reasons:

1. To benefit from owning it
2. To be able to sell it later

The value of a financial asset (a **security**) depends on:

3. The <u>size of the cash flows</u> that it is expected to generate (dividends)
4. The <u>uncertainty</u> in one's forecasts of these cash flows.

Bonds

When the stream of payments from an asset is fixed, the price of the asset will be inversely related to the interest rate it yields.

Bond = A security that promises to pay fixed amount of money at specific intervals

- The risk of default on <u>government bonds</u> is usually negligible
- <u>Corporate bonds</u> are not risk-free: High risk of default → high interest rate demanded by investors → lower bond price

Stocks

Stocks (shares) = A claim on a part of the assets of a firm, and hence on its profits.

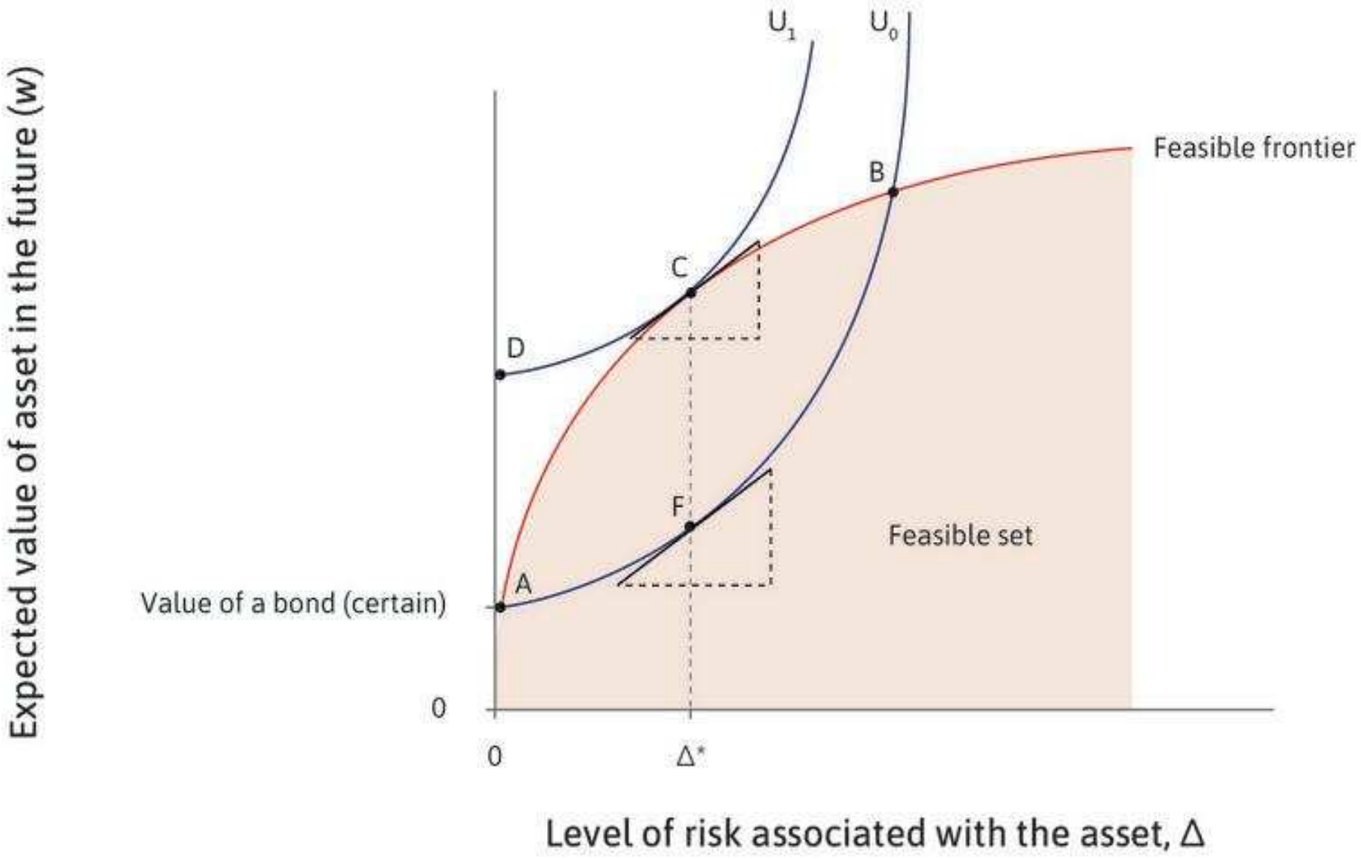
Stocks offer no specific promised stream of payments, and the time period over which payments will be made is not fixed.

Firms expected to generate greater net earnings will have higher valuations = higher share price.

The Trade-off between Risk and Return

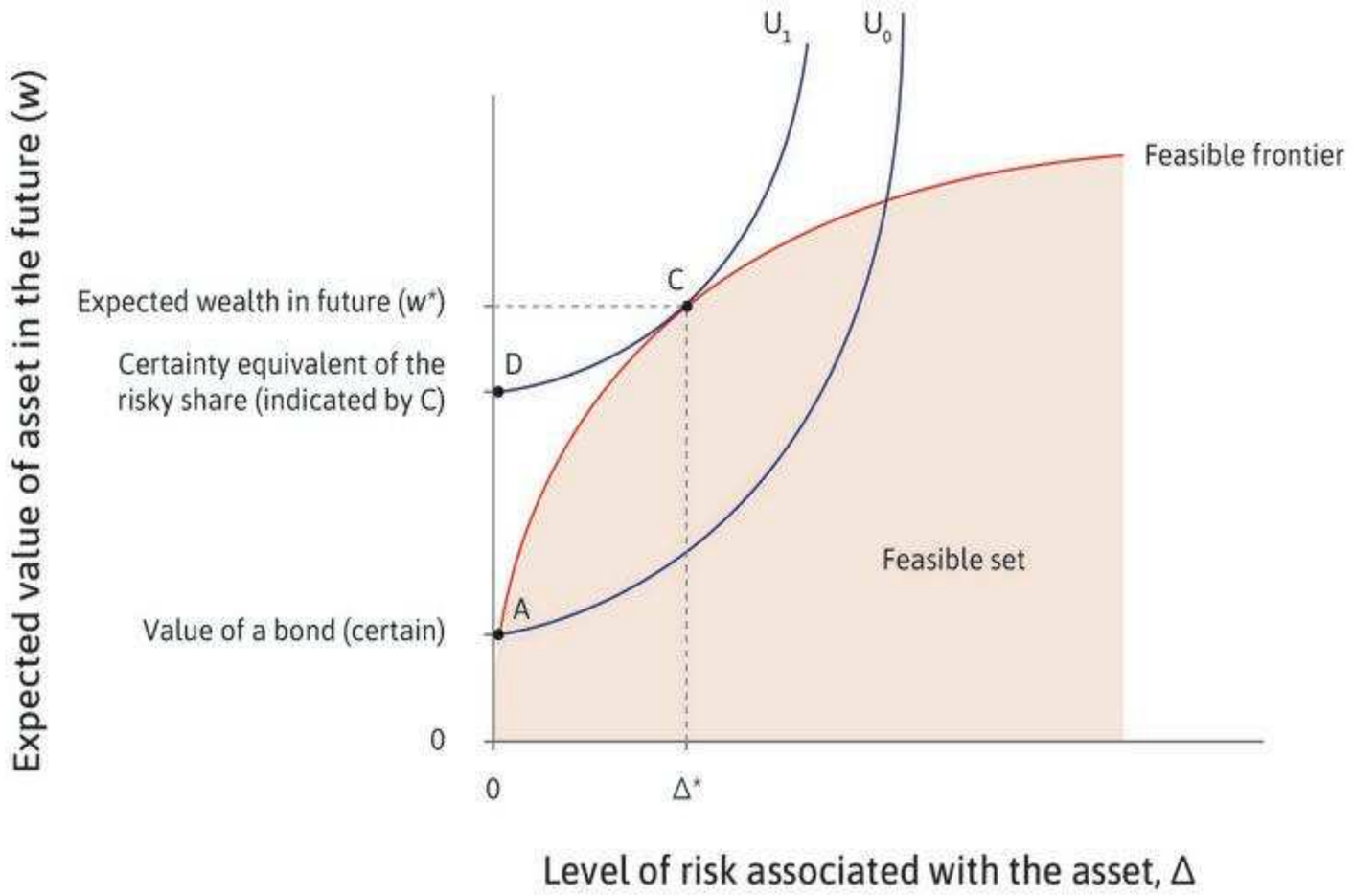
The slope of Ayesha's indifference curves is called the marginal rate of substitution between risk and expected value. A steep indifference curve means that taking on a given increase in risk would have to be compensated by a large increase in return.

At a given level of risk such as Δ*, she is less risk-averse when her expected wealth is higher: the slope is flatter at C than at F.



The trade-off between risk and return

The best Ayesha can do is to select point C, that is, a share with an expected value of *w** and a risk level of Δ*.



Asset Market Bubbles

Changes in asset prices are messages containing information about the firm's health, etc.

But if traders interpret a price increase as a sign of further price increases (**momentum trading**) the result can be <u>self-reinforcing cycles</u> of price increases (**bubbles**) followed by sudden price declines (**crashes**) when investor expectations shift

- <u>Resale value, ease of trading, availability of borrowing</u> may all make bubbles more likely

How a Bubble Forms

Good news about future profitability

↓

Demand rises and price increases

↓

Positive feedback: The price increase is interpreted as signal of future good news

↓

Further increases in demand

Prices and beliefs

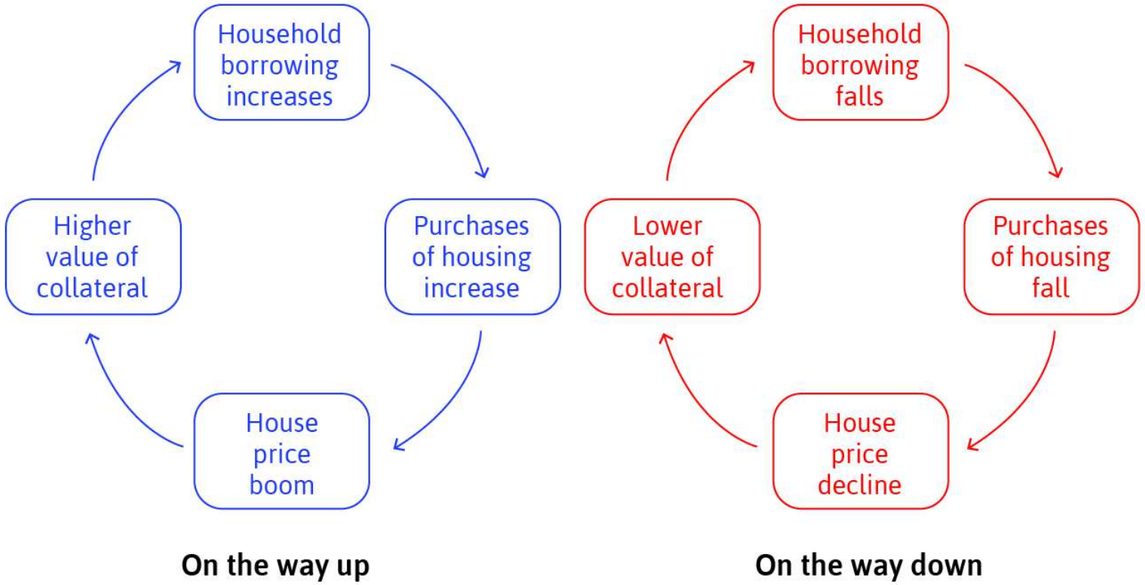
Beliefs dampen price rises

Beliefs amplify price rises: a bubble

Beliefs may dampen or amplify price rises

F. The Housing Bubble and the Global Financial Crisis

Housing boom and the financial accelerator



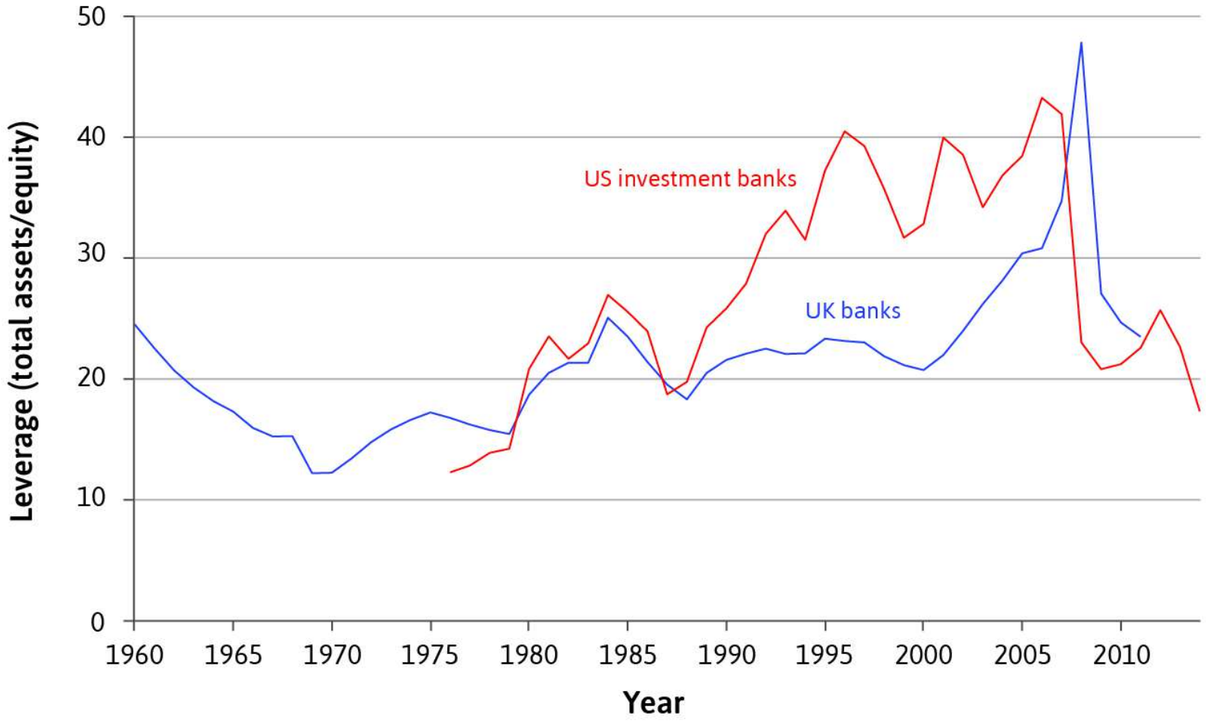
Buying a house via mortgage is a secured (collateralized) loan.

Financial accelerator: When house prices go up, so does the value of collateral, and households can borrow more.

This pushes up house prices further and sustains the bubble.

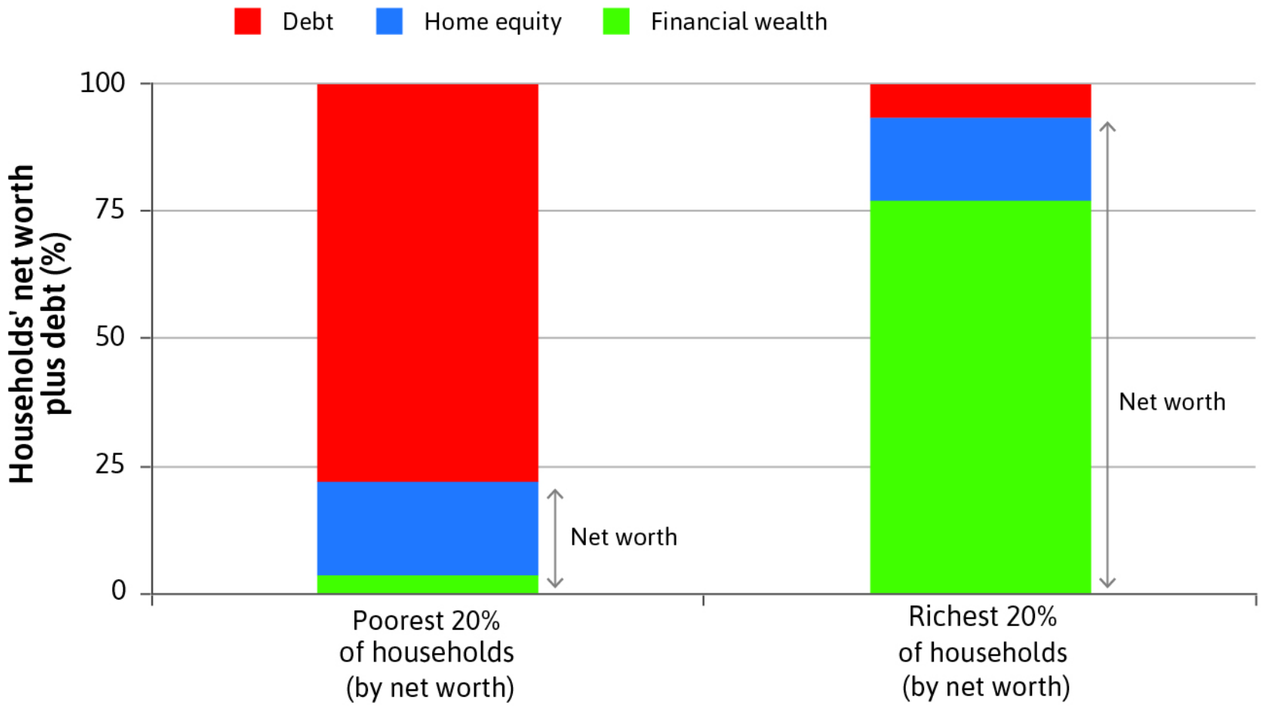
Financial deregulation

The combination of the great moderation, rising house prices, and the development of new, apparently less risky financial assets (**CDOs** and **MBSs**) made it profitable for banks to become highly leveraged.



Credit ratings agencies gave high ratings to many assets created from **subprime mortgages**.

Subprime borrowers

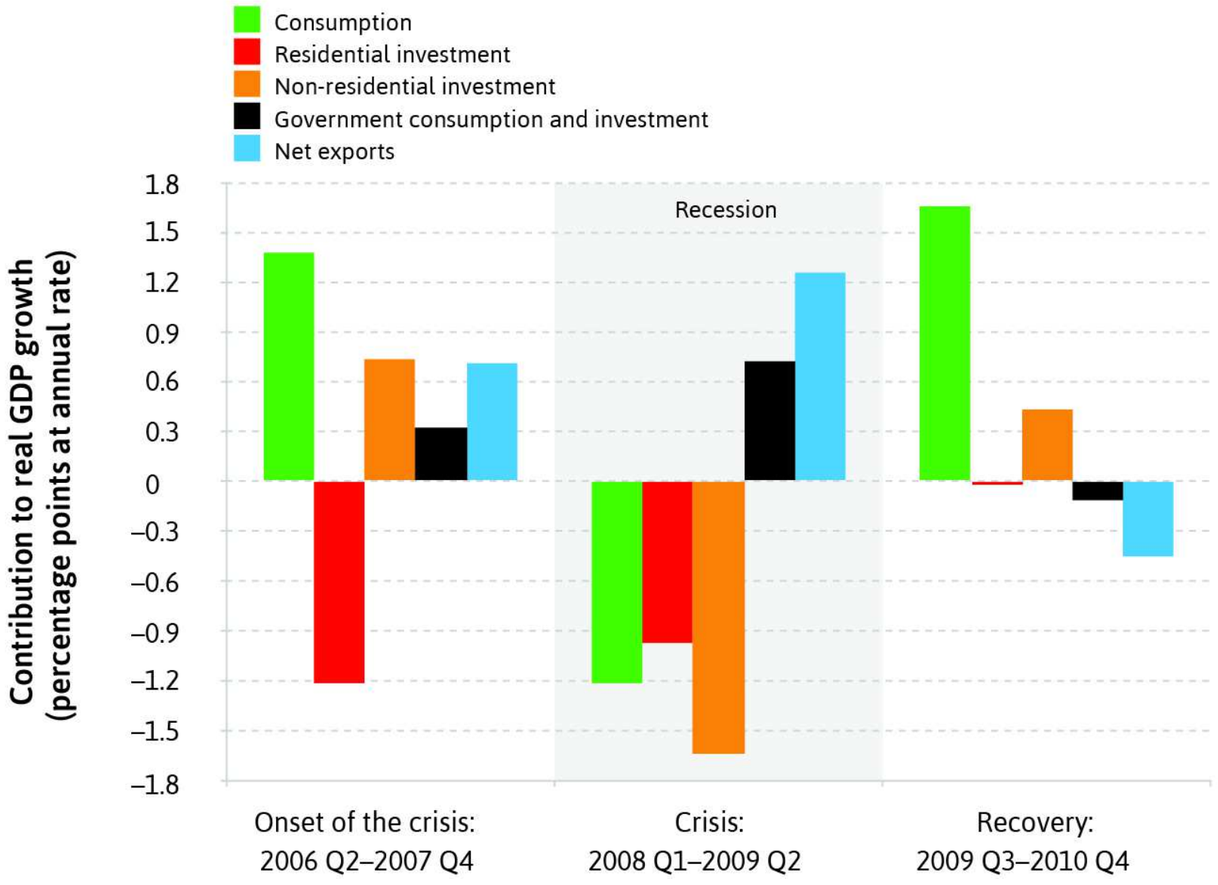


Poor households are normally only able to borrow when they have housing collateral. When house prices are expected to rise, the riskiness of home loans falls. Lenders ask for lower deposits, or even no deposit at all.

The financial crisis

The **great moderation** was ended by the global financial crisis, triggered by falling US house prices from 2007 onwards.

Despite the **bank bailouts** and stabilization policies, there followed a sustained global fall in aggregate output (the **great recession**).



The role of Banks in the Crisis

- Many banks were highly leveraged and at risk of **insolvency,** and their assets were of uncertain value

- Liquidity problems: To ensure payments on their liabilities banks became much less willing to lend, and wanted to sell assets

- **Fire sales** reinforced the fall in asset prices and worsened the insolvency of banks (positive feedback process)

- Governments had to rescue the banks